Swiss Job Market Trends and Salary Transparency Analysis

Course: Data Collection, Integration and Preprocessing (CIP)
Semester: FS26
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1 Introduction

Digital job platforms provide near real-time insight into labor market dynamics, including employer demand for skills, technologies, and experience levels. However, such information is fragmented across multiple online platforms and often lacks standardized structure, making systematic analysis difficult.

This project analyzes structural characteristics of the Swiss job market with a particular focus on salary transparency in online job advertisements. By collecting and processing job listings from Swiss job platforms and combining them with official labor market statistics, the project aims to uncover patterns in skill demand, regional variation, and salary disclosure practices.

The project emphasizes data acquisition, preprocessing, and integration using Python, demonstrating a complete data pipeline from raw web data to analytical insights.

1.1 Research Questions

The analysis is guided by three central research questions:

> RQ1: Which skills and technologies are most frequently requested in Swiss data-related job postings?
>
> RQ2: How do job requirements differ across regions, industries, roles, and seniority levels?
>
> RQ3: How prevalent is salary information in job postings, and does salary transparency vary by region or role?

2 Data Sources

2.1 Micro-Level Data

Micro-level data was collected by scraping job postings from Jobup.ch, one of Switzerland's primary online job platforms. The scraper targets five data-related roles: *data scientist*, *data analyst*, *machine learning engineer*, *data engineer*, and *AI engineer*. For each role, the collector issues paginated search queries and extracts structured metadata from both the search result pages and individual job detail pages, retrieving the job title, company name, location, posted date, contract type, workload percentage, and full job description. Duplicate listings are deduplicated using a SHA-1 hash of the source and URL, and the raw output is exported as a timestamped CSV to data/processed/.

2.2 Macro-Level Data

To contextualise the scraped postings within the broader Swiss labour market, official vacancy statistics were obtained from the Swiss Federal Statistical Office (BFS) via the PXWeb API. Two datasets were retrieved: one providing quarterly vacancy counts broken down by BFS major region, and one broken down by economic division under the NOGA classification. Both were filtered to the eight most recent quarters (2024Q1–2025Q4) to align with the micro-level data timeframe, yielding the regional and industry vacancy benchmarks used in the integration stage.

3 Data Acquisition & Preprocessing

Data acquisition was carried out using a custom Selenium-based collector targeting Jobup.ch. The scraper navigates paginated search result pages (SERP) for each configured role query and extracts job card metadata directly from the listing pages.

For each unique job posting URL discovered, the collector opens the corresponding detail page to retrieve richer structured information, including the full job description, workload range, and contract type, fields that are either absent or unreliable on the SERP cards alone.

The browser runs in headless mode via ChromeDriver, managed automatically using webdriver-manager. A polite sleep interval is applied between requests to avoid overloading the platform. Pagination stops either at the per-role page cap or when consecutive pages return no new listings. The pipeline yielded 743 unique job postings collected in early 2026 (2026Q1).

The raw dataset was then transformed through a structured preprocessing pipeline. Structural validation confirmed appropriate datatypes across all columns, verified that numeric fields such as workload percentages fell within expected ranges, and checked for outliers in extracted numeric variables.

Three further steps are worth highlighting. First, seniority was inferred using a two-pass classifier: an initial scan of job titles for explicit signals (e.g. *junior*, *senior*, *lead*), followed by a description-based fallback mapping years-of-experience mentions to seniority tiers. Despite both passes, 394 of 743 postings (53%) remained unclassified, reflecting genuine Swiss job posting norms rather than a pipeline failure.

Second, location was resolved through a 7-step priority chain covering canton codes, company-name lookups, multilingual canton name patterns, and city-to-canton dictionary matching, successfully assigning 644 of 743 postings to a BFS major region.

Third, skills were extracted using a predefined dictionary of 100+ technologies spanning programming languages, ML frameworks, cloud platforms, and data engineering tools, with all regex patterns using `\b` word boundaries to prevent false positives.

4 Data Integration

The cleaned micro-level dataset was merged with both BFS macro vacancy datasets to produce the final analysis-ready file (jobs_merged_final.csv), implemented in 04_micro_macro_integration.ipynb in two phases: a regional vacancy merge followed by an industry vacancy merge.

A key methodological challenge was a temporal mismatch between the two data sources. The vast majority of scraped postings date from 2026Q1, while BFS macro releases are published with a lag, the most recent available quarter at time of collection was 2025Q4. To preserve all job postings in the merge without producing null macro values, a merge_quarter proxy key was constructed that maps 2026Q1 to 2025Q4, stored separately from the true posting date throughout the dataset.

Since the scraped dataset contains no explicit NOGA industry codes, industry labels were inferred using a rule-based classifier applied to company names and job descriptions. Company names are checked first as the

more reliable signal, with a description-based fallback for postings from recruitment agencies, redirecting classification to the client industry rather than the agency's own sector.

The final dataset contains 743 rows and 22 columns with no duplicate records, combining micro-level job attributes with macro-level regional and industry vacancy benchmarks under a unified schema.

5 Analysis & Findings

5.1 RQ1: Skill Demand

Python, SQL, and cloud technologies, particularly Azure, emerge as the most frequently requested skills across all five roles, confirming their central importance in the Swiss data job market. The average number of required skills varies by role: Data Scientist and Data Engineer positions demand a broader set of technical competencies than Data Analyst roles, reflecting greater complexity and specialisation. These patterns hold consistently across regions and industries, suggesting a relatively standardised technical foundation across the market.

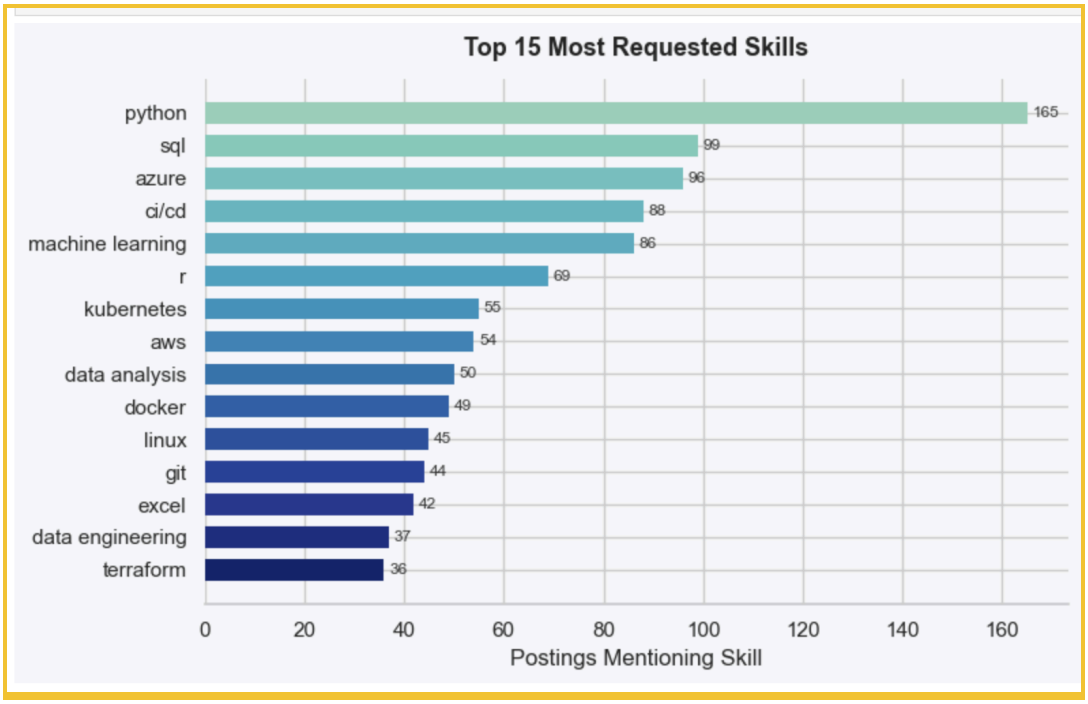

Figure 1. Top requested skills in Swiss data job postings. Python, SQL, and cloud technologies dominate across all roles.

5.2 RQ2: Regional and Industry Differences

Job postings are strongly geographically concentrated, with Zurich and the Lake Geneva Region together accounting for nearly 60% of postings with a resolved location. This reflects Switzerland's broader economic structure, where major urban centres act as innovation and employment hubs. Skill requirements vary moderately across regions, with major hubs demanding a wider range of technologies. This pattern is consistent across all roles, with Data Engineer positions being the most prevalent across regions.Across

industries, roles in professional services and finance tend to require more diverse skill sets than those in public administration or education sectors. The regional distribution of scraped postings aligns reasonably well with official BFS vacancy figures, supporting the representativeness of the dataset.

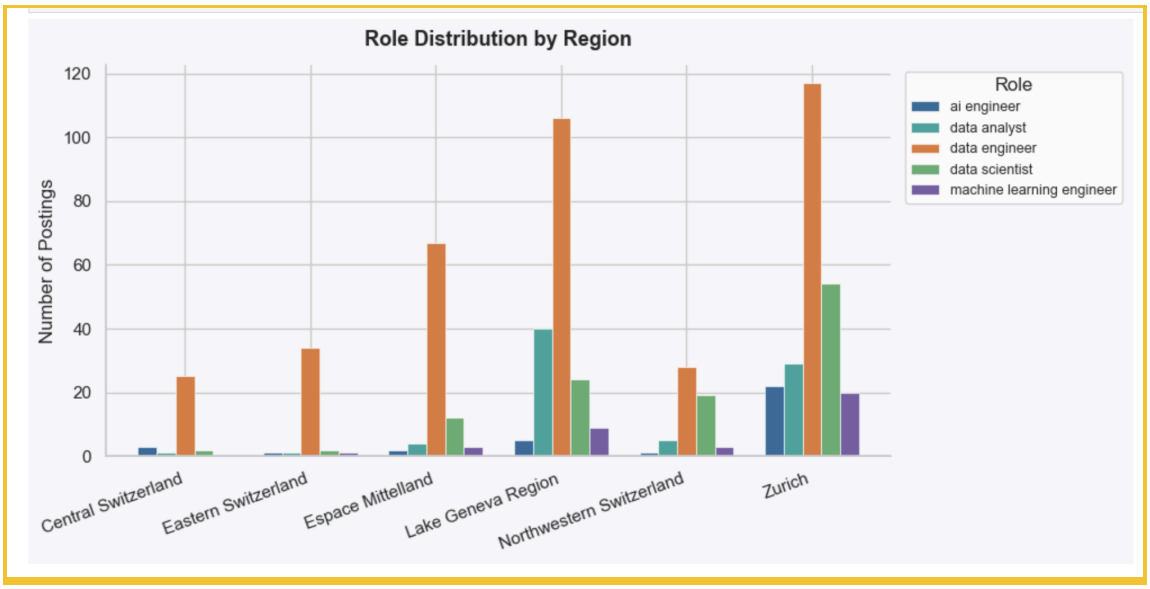

Figure 2. Distribution of data job roles across Swiss regions. Zurich and the Lake Geneva Region show the highest concentration of job postings across all roles.

5.3 RQ3: Salary Transparency

Salary transparency is extremely limited. Only 9 of 743 postings (1.2%) include explicit salary information, indicating a structural lack of compensation openness in the Swiss data job market that may disadvantage job seekers. Where salary data is present, technical roles such as Data Scientist show marginally higher disclosure rates, though the sample is too small to draw firm conclusions.

5.4 Interactive Dashboard

In addition to the static visualizations presented in this report, an interactive dashboard was developed using Streamlit to explore the processed dataset. The dashboard allows users to interactively examine skill demand, regional job distribution, and salary transparency in Swiss data-related job postings.

The full project code, including the scraping pipeline and dashboard implementation, is available in the GitHub repository: https://github.com/meleknurs/CIP_FS26_Group202

6 Conclusion & Discussion

This project analysed the Swiss data job market using 743 job postings scraped from Jobup.ch, enriched with official BFS vacancy statistics at the regional and industry level. The findings confirm that Data Engineer is the most in-demand role, with Python, SQL, and Azure as the core technologies requested across all positions. Geographic concentration is pronounced, with Zurich and the Lake Geneva Region dominating the

market. Most strikingly, salary transparency is near-absent, fewer than 2% of employers disclose compensation information, pointing to a structural openness gap in the Swiss labour market that warrants attention from both job seekers and policymakers.

The integration of macro-level BFS data strengthens the analysis by providing independent validation of the dataset's spatial patterns. The regional distribution of scraped postings broadly mirrors official vacancy figures, suggesting the dataset is a reasonable proxy for structural trends in the Swiss data and AI labour market despite covering only a single platform. The temporal proxy applied to bridge the 2026Q1 micro data with the 2025Q4 BFS release introduces only a minor mismatch and does not materially affect the regional or industry-level comparisons.

Future work could extend the dataset across a longer time horizon to enable trend analysis, incorporate additional platforms such as LinkedIn or Indeed to reduce platform bias, and apply more sophisticated NLP techniques for skill extraction and seniority classification — areas where the current rule-based approach reaches its limits.

7 Limitations

7.1 Data coverage

The dataset is restricted to job postings scraped from a single platform (Jobup.ch) over a short collection window in early 2026, representing a snapshot of the market rather than a longitudinal view.

7.2 Salary sparsity

Only 9 of 743 postings (1.2%) include explicit salary information. Salary-related findings should be interpreted as indicators of transparency rather than estimates of actual compensation levels in the Swiss data job market.

7.3 Seniority classification

The two-pass classifier relies on keyword signals in job titles and descriptions. 394 postings (53%) contain no detectable signal and remain unclassified, limiting the scope of seniority-based analysis.

7.4 Platform bias

Collecting exclusively from Jobup.ch may introduce platform-specific bias, as different job platforms attract different types of employers and postings. The dataset may therefore not fully represent the entire Swiss data job market.

7.5 SERP card data quality

Metadata extracted from search result page cards is often incomplete or imprecise, as Jobup dynamically renders these fields and their CSS selectors are not stable. The scraper mitigates this by preferring detail-page values over card-level values, but the pipeline remains sensitive to changes in Jobup's front-end structure.

8 Sources

8.1 Data Sources

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OpenAI. (2025). *ChatGPT* (GPT-4) [Large language model]. Used to support code development and debugging. https://openai.com

8.4 Project Repository

Özgü, M., Fuchs, E., & Rahimi, F. (2026). Swiss Job Market Trends and Salary Transparency Analysis (CIP Project). GitHub repository.

https://github.com/meleknurs/CIP_FS26_Group202